

AIM:

To develop an Android application that connects to the Internet and loads a web page using WebView with proper handling.

Procedure

- 1 Open Android Studio
 - . Create a new project (Empty Views Activity, Kotlin)
- 2 Add Internet permission in AndroidManifest.xml
 - . Design layout with WebView
- 3 Enable JavaScript in WebView
 - . Load a website using URL
- 4 Handle navigation and errors
 - . Run and test the application

AndroidManifest.xml

```
<uses-permission android:name="android.permission.INTERNET"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <WebView
        android:id="@+id/webView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.internetconnectapp
```

```
import android.os.Bundle
import android.webkit.WebResourceError
import android.webkit.WebResourceRequest
import android.webkit.WebView
import android.webkit.WebViewClient
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var webView: WebView
```

```

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    webView = findViewById(R.id.webView)

    webView.settings.javaScriptEnabled = true

    webView.webViewClient = object : WebViewClient() {

        override fun shouldOverrideUrlLoading(
            view: WebView?,
            request: WebResourceRequest?
        ): Boolean {
            view?.loadUrl(request?.url.toString())
            return true
        }

        override fun onReceivedError(
            view: WebView?,
            request: WebResourceRequest?,
            error: WebResourceError?
        ) {
            Toast.makeText(
                this@MainActivity,
                "Failed to load page",
                Toast.LENGTH_SHORT
            ).show()
        }
    }

    webView.loadUrl("https://www.google.com")
}

// Handle back navigation
override fun onBackPressed() {
    if (webView.canGoBack()) {
        webView.goBack()
    } else {
        super.onBackPressed()
    }
}
}

```

Result

- The application successfully connects to the Internet
- Loads a webpage inside WebView
- Handles navigation and errors

- Allows back navigation within the WebView

